Sahaj Bhadja

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Summary

Graduate Software Engineer with 2+ years building **scalable software**, AI products, distributed services, and cross-platform applications using Python, Java, TypeScript, REST, containers, automated testing, and AI-assisted engineering workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Architected a **multi-tenant platform** on Azure SQL and Blob Storage with row-level isolation and indexed queries, reducing dashboard latency from 1.2 seconds to 350 milliseconds for 500+ users.
- Reworked asynchronous **LLM orchestration** with FastAPI workers, explicit job-state transitions, and failure isolation, tripling throughput for 1K+ concurrent jobs while sustaining 99%+ reliability across 5K+ monthly runs.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Owned **AI grading workflows** from React interfaces through Flask APIs, integrating GPT-4o, Claude, Qwen-VL, and OCR to support 150+ instructors and 10K+ daily production requests reliably at scale.
- Designed concurrent ingestion for 1K+ files per batch using ThreadPoolExecutor, AWS RDS, and automated **CI/CD validation**, reducing peak-load feedback turnaround 60% while maintaining responsive user experiences globally.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a **Decision Transformer** for autonomous-driving offline reinforcement learning, processing 100K+ trajectory logs through reproducible Python evaluation pipelines and improving simulated planning performance 18% across research experiments.
- Architected a spatial-temporal encoder combining HD-map geometry, lane context, and ego trajectories with **multi-head attention**, improving closed-loop stability 15% while documenting benchmarks and failure modes for collaborators.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Simplified cross-platform integration services through **object-oriented design patterns**, cutting API latency 30%, technical debt 20%, and duplicated logic while improving maintainability across a reliable distributed data-sharing platform.
- Rebuilt AWS Glue and S3 **data pipelines** for incremental processing, improving efficiency 68%, then maintained 20+ automated test suites through Jenkins across Windows, Linux, and macOS environments.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a **React and Node.js** operations platform with REST APIs, WebSockets, permissions, and notifications, synchronizing 2M+ records across locations while reducing staged-release conflicts 33% during production rollouts.
- Owned eight modular services from implementation through **Docker and Jenkins** deployment, adding automated validation and production monitoring that reduced release time 50% and sustained 95%+ workflow reliability.

Skills

Languages: Python, Java, TypeScript, JavaScript, C++, SQL, HTML5, CSS3, Bash
Backend: FastAPI, Flask, Node.js, Spring Boot, REST APIs, gRPC, Microservices, Async Processing, Distributed Systems
Frontend & AI: React, Next.js, WebSockets, PyTorch, Transformers, LLM APIs, OCR, AI Coding Agents
Data & Cloud: PostgreSQL, MongoDB, Redis, Kafka, AWS, Azure, Docker, Kubernetes, Terraform
Quality & Tools: Git, Linux, PyTest, Jest, CI/CD, GitHub Actions, Jenkins, Sentry, Code Reviews

Projects

Adaptive Media Processing Platform | *Rust, Python, gRPC, Kafka, Kubernetes, Redis, Prometheus*

- Built a **distributed media pipeline** that selects Rust or Python workers by workload, streams tasks through Kafka and gRPC, and autoscaled Kubernetes services while preserving idempotency, observability, and fault isolation.

AI Product Experimentation Engine | *Python, FastAPI, React, PostgreSQL, Docker, GitHub Actions*

- Built an **AI-native experimentation system** that generates feature variants, assigns audiences, validates releases, and compares retention metrics, using automated tests and review gates to promote only reliable product changes.

PiXMod Live | *Next.js, TypeScript, PyTorch, Stable Diffusion, CLIP, MongoDB, Stripe*

- Built an AI-powered **consumer image platform** with segmentation, generation, restoration, semantic search, and subscription billing behind a responsive Next.js interface, increasing premium conversion 18% through streamlined creative workflows.